POLYTECHNIC UNIVERSITY OF THE PHILIPPINES

ARVACORE: AN IOT-BASED IRRIGATION MONITORING AND AUTOMATED WATER MANAGEMENT SYSTEM FOR RICE CULTIVATION

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by

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Interpretation of Weighted Means for User Acceptability Questionnaire.

Weighted Mean	Interpretation
1.00–1.80	Very Low Acceptability
1.81–2.60	Low Acceptability
2.61–3.40	Moderate Acceptability
3.41–4.20	High Acceptability
4.21–5.00	Very High Acceptability

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Diagram of Conceptual Framework

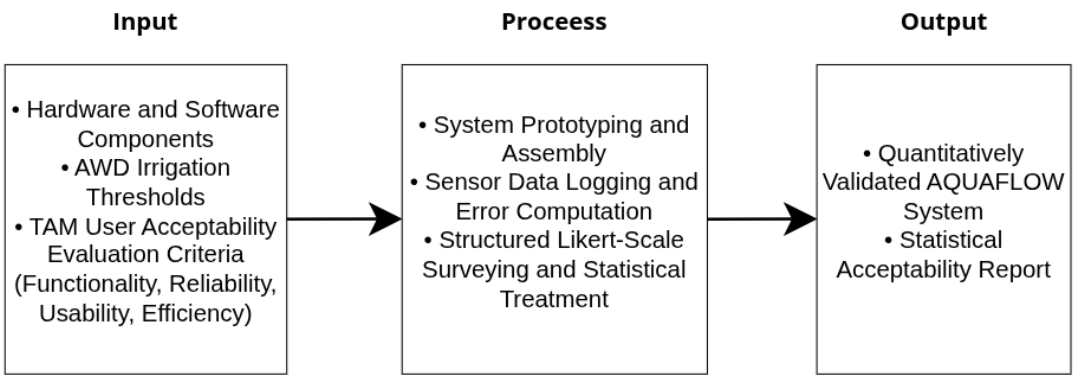


Figure 2.

ArvaCore Process Flowchart

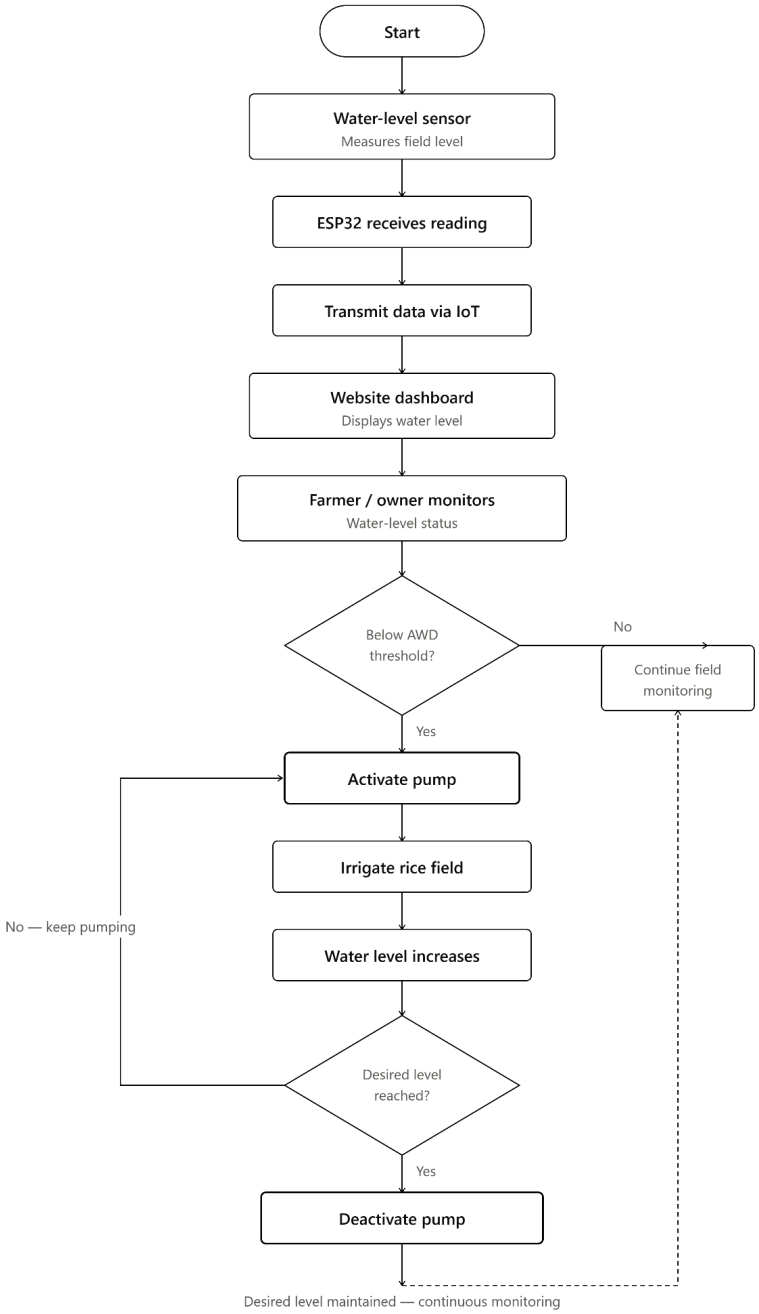
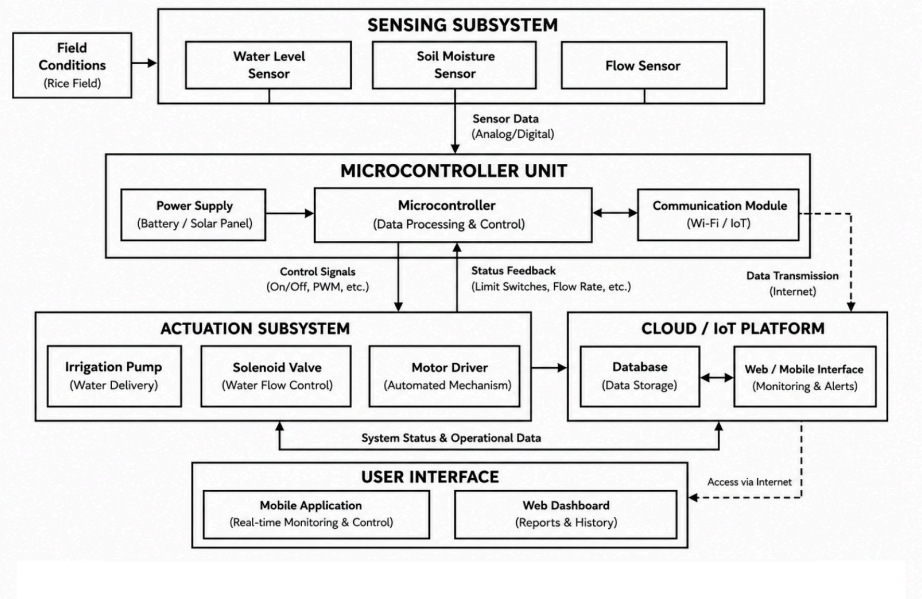


Figure 3.
ArvaCore System Block Diagram



Chapter 1

THE PROBLEM AND ITS SETTING

Introduction

Rice is a fundamental component of the Filipino diet and remains one of the country's most important agricultural commodities. However, producing it requires careful management of agricultural resources, particularly water, which is essential throughout rice cultivation. In irrigated rice farming, maintaining an appropriate amount of water in the field supports crop growth, while excessive or poorly timed irrigation can lead to inefficient water use. As demand for sustainable agricultural practices continues to rise, improving how irrigation water is monitored and managed has become an important consideration in rice cultivation. One approach to address this concern is precision irrigation management, a strategy in which water is delivered based on actual field conditions rather than fixed schedules or continuous flooding. By monitoring water levels and soil moisture in real time, farmers can avoid unnecessary irrigation and reduce water waste without compromising crop yield (International Rice Research Institute [IRRI], n.d.).

However, implementing precision irrigation requires farmers to have reliable information about their fields' water conditions. This creates an opportunity to apply the Internet of Things (IoT), which refers to interconnected physical devices equipped with sensing, processing, and communication capabilities that enable them to collect and

exchange data (National Institute of Standards and Technology [NIST], n.d.). In agriculture, IoT technologies can monitor field and environmental conditions, including soil moisture, water levels, temperature, and other parameters relevant to crop production. Through continuous or periodic data collection, IoT-based systems can provide farmers with timely information to support irrigation decisions and improve agricultural resource management (Chamara et al., 2022). Integrating IoT monitoring with AWD therefore offers an opportunity to make water-management practices more informed and responsive to actual field conditions.

Despite the potential of precision irrigation to improve water-use efficiency, implementing it under real farming conditions presents several challenges. Factors such as farmers' knowledge and access to information, irrigation conditions, institutional arrangements, and economic considerations have influenced the adoption of improved irrigation practices among rice farmers in the Philippines (Lampayan et al., 2015). In addition, effective irrigation management requires regular observation of field water levels to determine when to apply water. The IRRI recommends the use of a field water tube or a similar monitoring method to determine the depth of water relative to the soil surface and guide irrigation decisions (IRRI, n.d.). For farmers who depend on shared irrigation systems, this requirement is accompanied by concerns about irrigation water availability, coordination among users, and the practicality of consistently monitoring field conditions.

These concerns are evident in the rice-farming activities of Barangay Silang, Lopez, Quezon, where farmers rely on irrigation to sustain rice cultivation. The area’s

agricultural activities are supported by irrigation infrastructure and management arrangements through the National Irrigation Administration (NIA) and organised irrigation groups (National Irrigation Administration [NIA], n.d.). In this setting, local farming experience highlights several irrigation management challenges. The local irrigation organization provides machinery and equipment for irrigation operations, making rising equipment costs a concern. Farmers also monitor field water conditions largely through manual observation, requiring them to inspect their fields to determine whether irrigation is necessary. Furthermore, the use and distribution of irrigation water requires coordination among farmers who depend on the same irrigation system. These conditions show that effective irrigation management depends not only on water availability and infrastructure but also on access to appropriate equipment and timely information about field conditions.

These challenges are particularly relevant to implementing precise irrigation management. Although monitoring-based irrigation can reduce unnecessary water use, its effectiveness depends on accurately determining when a field requires water and when irrigation should be stopped.. Reliance on manual observation can make this process more demanding, especially when farmers need to monitor field conditions repeatedly throughout the cultivation period. At the same time, additional irrigation equipment may introduce financial and operational concerns. IoT-based monitoring offers an alternative way to collect field-condition data through sensors and communication technologies, allowing information to be collected and accessed without relying solely on direct, repeated field observation. Recent research has shown the potential of IoT-based sensing systems to support irrigation management by

providing information that helps determine irrigation requirements and improve water-use efficiency (Abdelmoneim et al., 2025; Chamara et al., 2022).

In response, the proponents propose ArvaCore: An IoT-Based Irrigation Monitoring and Water Management System for Rice Cultivation. The proposed system will integrate IoT-based sensing and monitoring capabilities to provide timely information on relevant field water conditions and support irrigation-related decision-making. ArvaCore is intended to complement existing irrigation practices by providing a technological means of monitoring field conditions and managing irrigation information rather than replacing existing irrigation infrastructure. By integrating IoT technology and water-management principles, the study aims to develop a practical monitoring and management system that supports more efficient irrigation in rice cultivation.

Theoretical Framework

The study is primarily anchored on the Technology Acceptance Model (TAM) developed by Davis (1989). TAM explains technology acceptance through two primary constructs: Perceived Usefulness (PU) and Perceived Ease of Use (PEOU). The model has been widely applied in agricultural technology research to examine farmers' acceptance of digital, IoT-based, and smart-farming technologies. Recent studies have demonstrated the relevance of PU and PEOU in understanding farmers' acceptance of agricultural technologies, particularly those designed to improve productivity, resource efficiency, information access, and farm management (Arrang et al., 2025; Thomas et al., 2023).

Perceived Usefulness (PU) refers to the extent to which users believe that using a particular technology would improve their performance in carrying out their activities (Davis, 1989). In the context of ArvaCore, PU refers to the extent to which rice farmers and individuals involved in irrigation management in Lopez, Quezon, perceive the system as useful for improving irrigation practices and water management. ArvaCore is designed to provide timely information on water levels and soil conditions, support irrigation decisions, monitor water consumption, and facilitate automated irrigation-related processes. These capabilities are intended to help users manage irrigation more effectively while reducing unnecessary water use and manual effort. This construct is operationalized through the evaluation dimensions of Functionality and Efficiency. Functionality assesses the extent to which ArvaCore performs its intended monitoring, information, and irrigation-control functions, while Efficiency assesses the extent to which the system helps users accomplish irrigation-related activities with minimal effort and resource consumption. These dimensions collectively reflect users' perceptions of the practical benefits and performance improvements provided by ArvaCore. Studies on agricultural technology adoption likewise indicate that perceived practical benefits, resource efficiency, productivity improvements, and access to useful information contribute to farmers' acceptance of digital agricultural technologies (Thomas et al., 2023; Arrang et al., 2025).

Perceived Ease of Use (PEOU) refers to the extent to which users believe that using a technology requires minimal effort (Davis, 1989). For ArvaCore, PEOU concerns the convenience, simplicity, and effort required to interact with and utilize the

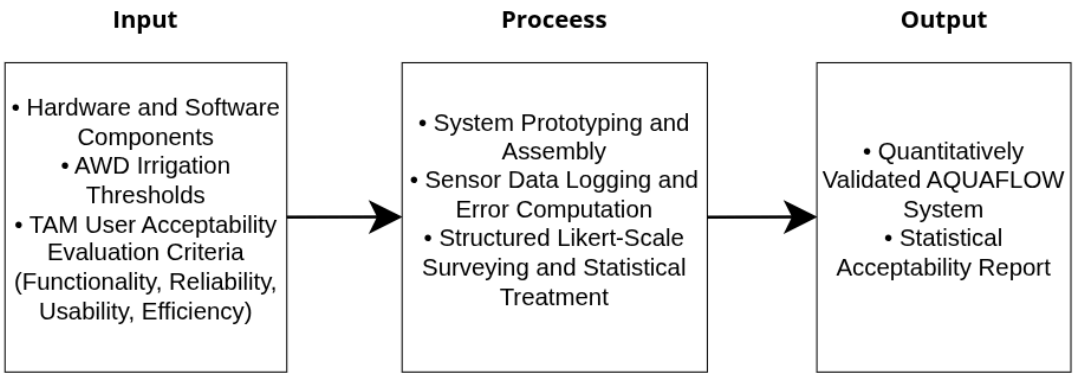
system for irrigation-related activities. ArvaCore is intended to reduce reliance on repeated manual field observation by providing remotely accessible information and automating selected irrigation processes. This construct is operationalized through the evaluation dimensions of Usability and Reliability. Usability assesses the extent to which users can understand, learn, and operate ArvaCore with ease, while Reliability assesses the consistency and dependability of the system in performing its intended functions. Together, these dimensions reflect users' perception of ArvaCore as a convenient, manageable, and dependable technology for irrigation management. The importance of ease of use, accessibility, and low user effort is likewise supported by research on the acceptance of digital agricultural technologies among farmers (Thomas et al., 2023).

The Internet of Things (IoT) framework further supports the study by providing the technological foundation for ArvaCore. IoT-based agricultural systems integrate sensors, devices, communication technologies, and software to collect, transmit, and process field data for monitoring and decision-making. Its application in agriculture enables the collection of real-time environmental and operational data and supports more efficient farm management (Bulut & Wu, 2022). In ArvaCore, the IoT framework is implemented through water-level, soil-moisture, temperature, humidity, and water-flow sensors, together with communication technologies and automated irrigation components. These technologies enable the system to gather relevant field conditions, transmit information, and support automated irrigation management.

Within the framework of the study, TAM provides the basis for evaluating the acceptability of ArvaCore, while the IoT framework explains the technological foundation through which the system performs its intended functions. The TAM constructs are operationalized by assessing Functionality and Efficiency under Perceived Usefulness, and Usability and Reliability under Perceived Ease of Use. This operationalization allows the study to examine both the practical benefits that users derive from ArvaCore and the degree of convenience and effort required to use it. The resulting assessments provide the basis for determining the acceptability and practical applicability of ArvaCore among rice farmers and individuals involved in irrigation management in Lopez, Quezon. Recent empirical and meta-analytic evidence on agricultural technology acceptance further supports the use of PU and PEOU as important constructs for understanding farmers' acceptance of digital and smart agricultural technologies (Arrang et al., 2025; Thomas et al., 2023).

Conceptual Framework

Figure 1.
Diagram of Conceptual Framework



The system's development and evaluation follow a structured three-phase methodological workflow.

The **Input Phase** establishes the study's foundational requirements and evaluation metrics, including the necessary hardware and software components, the specific Alternate Wetting and Drying (AWD) irrigation thresholds, and the Technology Acceptance Model (TAM) user acceptability criteria structured around functionality, reliability, usability, and efficiency.

These inputs are operationalized during the **Process Phase**, which involves physical system prototyping and assembly, rigorous sensor data logging paired with error computation, and the deployment of structured Likert-scale surveys subjected to appropriate statistical treatment.

Finally, the **Output Phase** delivers the fully realized, quantitatively validated system accompanied by a comprehensive statistical acceptability report confirming its operational and user-centric performance.

Statement of the Problem

This study aims to design, develop, and evaluate ArvaCore: An IoT-Based Alternate Wetting and Drying (AWD) Irrigation Monitoring and Water Management System for Rice Cultivation. Specifically, this study seeks to answer the following questions:

1. What are the current irrigation practices and challenges encountered by rice farmers in Barangay Silang, Lopez, Quezon?
2. What are the stages involved in the development of the ArvaCore system using the prototyping method?
3. How effective is the developed hardware of the ArvaCore system in monitoring field conditions and managing irrigation operations in terms of:
    3.1. Water-level monitoring accuracy;
    3.2. Soil-moisture monitoring accuracy;
    3.3. Irrigation activation accuracy;
    3.4. Irrigation deactivation accuracy;
    3.5. Irrigation response time;
    3.6. Water consumption monitoring?
4. What is the level of acceptability of the developed ArvaCore system as evaluated by rice farmers and agricultural practitioners in terms of:
    4.1. Functionality;

4.2. Reliability;

4.3. Usability;

4.4. Efficiency?

5. What is the expected output of the study based on the developed and evaluated ArvaCore system?

Scope and Limitations of the Study

The study covers the development and description of the ArvaCore prototype, an IoT-based irrigation monitoring and water management system specifically designed to support Alternate Wetting and Drying (AWD) practices in rice cultivation. The scope encompasses the technical design and assembly of four distributed sensor nodes for monitoring water level, soil moisture, temperature, and humidity; the implementation of a wireless communication network and cloud-based application server for data processing, storage, and remote monitoring; and the integration of a centralized automated irrigation control mechanism with a flow sensor to verify water delivery. The system's technical performance will be quantitatively described through measures of sensor accuracy, irrigation activation and deactivation precision, system response time, and the reliability of its notification and data logging capabilities. Furthermore, the study describes user acceptability using a structured questionnaire grounded in the Technology Acceptance Model (TAM), administered to 346 rice farmers and agricultural practitioners from the Municipality of Lopez, Quezon, to assess their perceptions of the system's functionality, reliability, usability, and efficiency. The geographic scope of the study is confined to the Municipality of Lopez, Quezon, with field testing of the ArvaCore prototype conducted in Barangay Silang.

However, the sample population is drawn from rice farmers and agricultural practitioners across the municipality to account for individuals who may cultivate land within the study area or possess relevant experience with rice irrigation practices under similar agricultural conditions.

The study acknowledges several limitations that define its boundaries. The technical evaluation of the ArvaCore prototype will be conducted under controlled conditions and within a specific field environment in Barangay Silang, Lopez, Quezon; therefore, the performance results may not generalize to other field conditions, soil types, or climatic environments without further testing and calibration. Although the user acceptability assessment includes respondents from across the Municipality of Lopez, their responses are based on their perceptions and experiences and may not fully represent the conditions of every rice-farming area within the municipality. The assessment also relies on self-reported data collected through a survey questionnaire, which is subject to biases inherent in such instruments, including social desirability bias and personal subjective interpretation of the questions. The study only evaluates the system based on the specific sensor models and hardware components selected for the prototype, and its long-term operational stability and durability may vary with different configurations or under prolonged use. The system is specifically designed to support AWD irrigation and may not be directly applicable to other irrigation methods or crops without significant modifications. The study is cross-sectional, capturing user perceptions at a single point in time, and does not examine long-term adoption behavior or the system's sustained impact on water savings and yield over multiple cropping seasons. Additionally, the study does not include a full economic cost-benefit

analysis of implementing the system, focusing instead on its technical and user-centered performance.

Significance of the Study

The successful development and evaluation of ArvaCore will provide significant benefits to the following:

Rice Farmers of Barangay Silang, Lopez, Quezon - Farmers will directly benefit from the system's real-time monitoring and automated irrigation capabilities. ArvaCore reduces the need for frequent manual field inspections by providing timely, accessible information on water levels and soil moisture. This supports more efficient water management, reduces labor, and supports proper implementation of Alternate Wetting and Drying (AWD), potentially lowering water consumption and irrigation costs without compromising crop yield.

Irrigation Groups and the National Irrigation Administration (NIA) - The system provides a data-driven tool for managing water distribution. The monitoring dashboard and historical data logging capabilities allow for better coordination among farmers and informed planning of irrigation schedules, addressing challenges related to water allocation and equipment management.

Agricultural Practitioners and Advisors - The proposed system serves as a practical reference for integrating IoT technology into rice farming. Documenting its development and performance can guide extension workers and agricultural

practitioners in promoting technology-supported water management strategies to farmers.

Future Researchers - The study contributes a working model of an IoT-based AWD monitoring system. The documented technical performance data, including sensor accuracy and system response times, provides a foundation for further studies. Researchers can use these findings to improve upon the system, adapt it to other crops or regions, or investigate additional variables in smart farming, such as integrating weather forecasting or developing more advanced predictive irrigation models.

Definition of Terms

Alternate Wetting and Drying (AWD):

Field Monitoring Node:

Conceptual - A deployed unit consisting of sensors and processing components that collect environmental or field data from a specific location.

Operational - In this study, a field monitoring node refers to one of four modular units deployed at designated locations within the rice field, each consisting of an ESP32 microcontroller, a water-level sensor, and a wireless communication module, operating independently to collect water-level measurements from its location.

Gateway:

Conceptual - A network device that connects and translates data between different communication protocols or networks.

Operational - In this study, the gateway refers to the primary communication intermediary that receives sensor data from the four ESP32-based field monitoring nodes and forwards the collected information through an Internet connection to the cloud application server, establishing the communication link between the agricultural field and the cloud infrastructure.

Internet of Things (IoT):

Conceptual - A network of interconnected physical devices equipped with sensing, processing, and communication capabilities that enable them to collect and exchange data.

Operational - In this study, IoT refers to the technological framework underlying ArvaCore, comprising interconnected water-level sensors, flow sensors, microcontrollers, wireless communication modules, and cloud-based data processing that collect, transmit, and process field water condition data to support irrigation monitoring and management.

Irrigation:

Conceptual - The artificial application of water to soil or land to assist in the growth of agricultural crops, typically through controlled systems such as canals, pipes, sprinklers, or pumps.

Operational - In this study, irrigation refers to the controlled delivery of water to rice paddies through the ArvaCore system's centralized irrigation mechanism, which automatically activates or deactivates a water pump based on real-time water-level

measurements and predefined Alternate Wetting and Drying thresholds to maintain optimal field conditions for rice cultivation.

Ultrasonic Water-Level Sensor:

Conceptual - A sensor that measures distance by emitting ultrasonic waves and calculating the time taken for the reflected waves to return.

Operational - In this study, the ultrasonic water-level sensor is the primary sensing component at each field monitoring node, measuring the current water level at designated monitoring locations, with its accuracy assessed by comparing readings against manual field measurements taken with a ruler or measuring stick.

Water-Flow Sensor:

Conceptual - A device that measures the rate or volume of fluid passing through a pipeline.

Operational - In this study, the water-flow sensor is installed in the centralized irrigation line to measure the amount and continuity of water delivered to the rice field during irrigation, providing flow-rate measurements that allow the system to verify water delivery when the pump is activated and to monitor irrigation water consumption throughout the process.

Wireless Communication Network:

Conceptual - A system of interconnected devices that transmit data without physical cables using radio frequency or other wireless technologies.

Operational - In this study, the wireless communication network refers to the data-transmission infrastructure connecting the four ESP32-based field monitoring nodes to the gateway, enabling sensor data collected in the agricultural field to be delivered to the cloud application server for processing and monitoring.

Wireless Communication Network:

Conceptual - A system of interconnected devices that transmit data without physical cables using radio frequency or other wireless technologies.

Operational - In this study, the wireless communication network refers to the data transmission infrastructure connecting the four ESP32-based field monitoring nodes to the gateway, enabling sensor data collected within the agricultural field to be delivered to the cloud application server for processing and monitoring.

Chapter 2

REVIEW OF LITERATURE AND STUDIES

Conventional Irrigation System in Rice Cultivation

Maintaining enough water in the field through frequent or continuous irrigation is a common component of conventional irrigation in rice farming. Paddy production uses a significant amount of irrigation water in some agricultural areas, and rice requires a lot of water. Sahoo and Associates (2024) highlighted that extensive irrigation techniques can place significant strain on available water resources, especially in areas where groundwater and other freshwater sources are crucial to agriculture. When irrigation is applied without considering actual soil moisture conditions, conventional irrigation can also lead to inefficient water use. Therefore, the need for better water management and delivery systems has grown, especially as water scarcity and shifting rainfall patterns affect agricultural production.

Traditional surface irrigation is still widely adopted since it tends to be simpler and more affordable for farmers to use compared with certain modern irrigation systems. As Arouna et al. (2023) note, surface irrigation techniques such as basin, border, and furrow methods are frequently used in rice farming, though they can suffer from low irrigation efficiency if water management and system design are inadequate. Continuous flooding can also lead to waterlogging and higher water use, creating challenges in water-scarce regions. These drawbacks have prompted researchers to

explore irrigation approaches that sustain rice yields while reducing unnecessary water use.

The drawbacks of traditional irrigation highlight the need to improve how water is supplied and managed in paddy fields. Arouna et al. (2023) pointed out that water-saving technologies can boost water-use efficiency relative to conventional irrigation, though their uptake may be influenced by cost, technical demands, and farmers' comfort with established methods. Thus, conventional irrigation serves as a crucial benchmark for assessing alternative systems, since its water usage and management techniques can be contrasted with more efficient options. Recognizing these traits is essential to designing an irrigation system that minimizes water loss without compromising rice yields.

Alternate Wetting and Drying (AWD) Irrigation in Rice Cultivation

Alternate Wetting and Drying (AWD) is an irrigation method that saves water by letting rice paddies dry to a certain point before reflooding them. Bo et al. (2024) examined 1,187 paired field observations and found that AWD's impact depends heavily on the lowest soil water potential reached during the dry phase. Their results show that soil water potential-based AWD can boost water productivity across a large share of global irrigated rice areas while also supporting sustainable rice farming.

Bwire et al. (2024) characterized AWD as a climate-smart water management approach that can enhance both water and crop productivity while lowering methane emissions. Still, how effective AWD is depends on variables like soil type, how often irrigation occurs, and careful tracking of water levels in the field. The author also noted that frequent manual monitoring is a major obstacle to implementing AWD. They

recommended using IoT-based smart sensors as a possible way to improve precision water management in rice fields.

Haque et al. (2021) explored AWD combined with organic soil additives such as rice husk biochar, oil palm biochar, and compost. The research found that ADW with biochar enhanced water-use efficiency, and rice yields with some organic additives were similar under AWD and continuous flooding. These outcomes suggest that pairing AWD with suitable soil amendments can enhance water-use efficiency without necessarily cutting rice production. This reinforces the idea that AWD could serve as a sustainable irrigation approach for growing rice.

Water and Soil Monitoring for Rice Irrigation

Close monitoring of water levels is crucial for managing irrigation in rice paddies, especially under water-saving irrigation strategies. Dela Cruz et al. (2022) developed a low-cost wireless sensor for real-time water-level monitoring in lowland rice fields. The sensor matched manual measurements well, showing relatively low errors in lab, simulated field, and actual field tests. Their results show that wireless sensing technology can offer an affordable and dependable way to monitor field water levels and support efficient irrigation management.

Recent progress has also brought about more automated methods for monitoring water levels. Hasan et al.(2025) built a computer-vision system using an attention-based ConvNeXt architecture to measure water height in AWD rice farming. The system achieved an R^2 of 0.989 and an RMSE of 0.523 cm, demonstrating high accuracy in estimating water levels. This method illustrates how automated monitoring

can reduce the drawbacks of manual measurements and aid real-time irrigation decisions.

Tracking soil moisture alongside water level can provide a more comprehensive basis for irrigation management. Amin et al. (2025) explored IoT-based soil moisture monitoring under AWD irrigation and discovered that continuous monitoring helped assess soil water conditions, root growth, and crop yield. Their work showed that soil amendments and water management practices can affect soil moisture availability and rice growth. These results highlight the need to incorporate soil moisture monitoring into smart irrigation systems to enhance water-use management in rice farming.

Temperature and Humidity Monitoring in Agricultural Fields

Ansari et al. (2023) created an Internet of Things (IoT)-based solution to track environmental temperature and humidity in agricultural farms, utilizing a DHT22 sensor, TTGO T-Beam, Wi-Fi, and LoRaWAN technology. The research pointed out that temperature and humidity are key environmental factors influencing plant growth and water conditions, so continuous monitoring of these variables is valuable in agricultural production. The proposed system collected temperature and humidity readings via the DHT22 sensor and sent the data through the TTGO T-Beam to the cloud using Firebase, enabling remote access through smartphones and web applications. In addition, the system employed threshold values to trigger a buzzer and alert users whenever environmental conditions reached critical levels. In testing, the system effectively transmitted sensor data and offered real-time monitoring and alerts, showing its capability for remote agricultural environmental monitoring. These results support adopting IoT-based sensor systems in agriculture, especially for ongoing

environmental monitoring and delivering timely data that can help farmers respond to temperature and humidity changes.

Anuar et al. (2024) showcased the use of IoT technology for real-time temperature and humidity monitoring with a DHT11 sensor, NodeMCU ESP8266, and Arduino IoT Cloud. The system sent environmental data to an online dashboard and allowed mobile access to the recorded measurements, while alerts were produced when conditions fell outside preset ranges. Although the prototype was tested indoors, its ability to continuously track environmental conditions shows how IoT-based sensing can be applied in agricultural contexts. This supports the use of sensor-based monitoring for systems like ArvaCore, where real-time field data can support more efficient irrigation management.

Wang (2024) introduced a multi-node temperature monitoring approach for digital agricultural planting, relying on a wireless sensor network to track agricultural temperature conditions in real time. The system captured temperature data from multiple monitoring nodes and relayed it to a central computer for analysis. The system also used a Bayesian classifier to detect abnormal temperature conditions and issue warnings to administrators. The study achieved abnormal-temperature monitoring accuracy exceeding 96%, illustrating the potential of wireless sensor networks for precise environmental monitoring and prompt detection of irregularities. This research is relevant to the current ArvaCore system because it shows how sensor networks can continuously gather agricultural environmental data and provide timely alerts. Likewise, ArvaCore applies sensor-based monitoring to rice farming, especially to track field water conditions and guide suitable irrigation decisions under the Alternate Wetting and Drying (AWD) approach.

Sensor-based systems can also support environmental monitoring by providing farmers with real-time field information. Rathod et al. (2020) built an IoT-based agricultural monitoring stick that could gather temperature, humidity, and soil-moisture data through various sensors. Farmers could access the data remotely, enabling them to track field conditions without regularly inspecting the farm. The study highlights the potential of IoT and wireless sensors as practical tools for continuous environmental monitoring in agricultural fields.

IoT-Based Automated Irrigation and Water Management

The incorporation of the Internet of Things (IoT) into farming methods greatly improves water use efficiency and automated resource tracking (Obaideen et al., 2022). Through the use of linked sensing devices, automated irrigation systems track changing environmental factors like soil moisture, temperature, and surrounding humidity in real time. These data-driven systems enable precise water delivery based on crop needs, avoiding overwatering or under-watering. As a result, modern smart irrigation systems serve as crucial tools for sustainable farming, helping preserve freshwater supplies while supporting global sustainability targets (Obaideen et al., 2022).

Cloud-supported systems, telemetry data processing, and wireless sensor networks broaden the possibilities of real-time agricultural monitoring and decision-making (Et-taibi et al., 2024). Combining microcontrollers, telemetry sensors, and automated feedback loops enables farmers to automate field control operations and improve water delivery under changing environmental conditions.

Additionally, sophisticated methods that merge computational algorithms like fuzzy logic with genetic algorithms considerably reduce water loss and error margins compared with manual irrigation (Soambaton & Nugroho, 2025). As worldwide water shortages worsen, adopting scalable IoT systems and automated control methods offers a reliable way to maintain crop output while lowering operational resource costs (Et-taibi et al., 2024; Soambaton & Nugroho, 2025).

Real-Time Monitoring, Data Logging, and Notification System

Continuous observation and data recording are vital parts of smart farming systems because they enable real-time tracking of field conditions and storage for future reference. Nsoh et al. (2024) examined IoT-enabled automated irrigation systems and highlighted the role of sensors, connected devices, and data-processing technologies in managing irrigation in real time. Sensor data can help determine when irrigation is needed and support faster decisions based on actual field conditions. The review also underscored the value of combining multiple sensor systems to achieve more efficient, automated irrigation operations. These features can lessen the need for manual oversight and enhance water efficiency in farming.

Morchid et al. (2024) created an IoT-based smart irrigation management system that integrates embedded sensors, telemetry, cloud computing, and remote communication. The system collects and transmits agricultural data for monitoring while also supporting automated irrigation management. It also offers notifications, so users can stay informed about irrigation status without being on-site. This shows how IoT technology can combine real-time monitoring, data transfer, and user alerts in a single system. These capabilities are pertinent to ArvaCore, as the proposed system

likewise requires continuous monitoring, data recording, and prompt alerts to support AWD irrigation management.

Usability and Acceptability of Agricultural Technology

Romero-Riaño et al. (2022) investigated the usability of AgroRIEGO, an IoT-based decision support system designed to support irrigation management in farming. The system uses sensors to track soil moisture levels and delivers irrigation details through an online portal and Android app, enabling users to access and manage the system remotely. To assess the system, the researchers conducted a heuristic evaluation, focusing on usability aspects such as system visibility, flexibility, user control, error prevention, readability, and ease of use. The assessment revealed strengths in system visibility, real-world connection, design, colour, and readability, but weaknesses in user control, error support, latency reduction, and default settings. These outcomes highlight that agricultural tools should provide precise monitoring capabilities and also be straightforward to comprehend, navigate, and operate for their target users. This work matters to ArvaCore because the system's usability and acceptance play a key role in whether farmers can successfully use its monitoring, notification, and water-management features in real rice farming.

The usability and acceptance of agricultural technologies are key factors in deciding whether farmers and other users will embrace and keep using new systems. In their systematic review of technology acceptance in smart agriculture, Thomas et al. (2023) found that perceived effectiveness, productivity gains, information access, and training can foster technology acceptance. Conversely, complexity, infrastructure needs, extra workload, and failure to address real farming issues can hinder adoption.

These results indicate that agricultural technologies should be technically effective, practical, clear, and aligned with users' needs.

Zul Azlan et al. (2025) explored how IoT technologies are accepted in smart farming by applying an extended Technology Acceptance Model. Their results demonstrated that perceived usefulness and perceived ease of use strongly affect users' intentions to adopt IoT-based agricultural systems, and elements like system quality, information quality, experience, trust, and government support also shape acceptance. The study stresses that users tend to accept agricultural IoT technologies when they are useful, easy to use, dependable, and offer pertinent information. This underscores the importance of creating systems like ArvaCore with a simple interface and clear, understandable real-time irrigation data.

Synthesis of the Reviewed Literature and Studies

The reviewed literature and studies collectively highlight the growing necessity for more efficient and sustainable water management in rice farming. Conventional irrigation methods, especially continuous flooding and traditional surface irrigation, remain widely used because of their simplicity and low cost. Nevertheless, these approaches can lead to excessive water use, waterlogging, and poor water management when irrigation is not aligned with actual field conditions. These shortcomings underscore the need for alternative irrigation strategies that minimise unnecessary water use while sustaining rice yields.

Among the discussed alternatives, Alternate Wetting and Drying (AWD) has emerged as a promising water-saving irrigation technique for rice cultivation. Research indicates that AWD can enhance water productivity and promote sustainable rice

farming when drying and flooding cycles are carefully managed. It can also contribute to better water use efficiency and lower environmental impact. However, AWD success depends on variables such as soil type, irrigation scheduling, and precise monitoring of field water levels. A key obstacle noted in the literature is the requirement for regular manual checks, which can make proper implementation challenging for farmers.

The reviewed studies also emphasize the value of incorporating sensor based monitoring into irrigation management. Water-level sensors, soil-moisture monitoring systems, and automated computer-vision tools can deliver accurate, real-time data on field conditions. Similarly, IoT-based sensors and wireless networks enable continuous monitoring of temperature and humidity, as well as remote access to agricultural data. These insights suggest that integrating various environmental sensors can provide a stronger foundation for informed irrigation decisions.

The literature on IoT-driven automated irrigation also endorses the use of connected sensors, microcontrollers, wireless communication, cloud platforms, and automated feedback systems to enhance water management. These technologies facilitate real-time data collection, transmission, automated monitoring, and more precise irrigation control based on current environmental conditions. Research on data logging and notification systems further stresses the importance of storing field data and delivering timely alerts, enabling users to oversee irrigation remotely and respond quickly without being physically present. Additionally, the reviewed study indicates that technical performance alone is insufficient for the successful adoption of agricultural technologies. Usability and user acceptance are equally crucial, as people are more likely to adopt systems that are helpful, easy to use, dependable, and provide clear, relevant information. Elements such as perceived usefulness, perceived ease of use,

system quality, information quality, and simplicity can shape acceptance of IoT-based agricultural technologies.

Overall, the literature and studies confirmed that AWD is a feasible method for enhancing water use efficiency in rice farming, and that IoT-based monitoring technologies can overcome the challenges associated with manual observation and irrigation management. However, the reviewed studies generally concentrate on specific elements, including water level monitoring, soil moisture sensing, environmental monitoring, automated irrigation, data logging, or technology acceptance. This creates an opportunity to integrate these features into a single system tailored to support AWD irrigation in rice farming. Thus, the proposed ArvaCore: An IoT-Based Alternate Wetting and Drying (AWD) Irrigation Monitoring and Water Management System for Rice Cultivation seeks to fill this gap by combining real-time tracking of relevant field conditions, data logging, alerts, and water-management assistance into one platform. By providing timely, easily accessible information on rice field conditions, ArvaCore aims to reduce reliance on manual checks and help users make better-informed irrigation decisions. Insights from the reviewed literature lay the groundwork for developing the system and underscore the need to assess not only its monitoring and water-management functions but also its usability and acceptance among target users.

Chapter 3

METHODOLOGY

Research Design

The study employs a quantitative research design, as it collects and analyzes numerical data to evaluate both the technical performance of the ArvaCore prototype and its acceptability among intended users. More specifically, this study uses a descriptive-developmental research design, which is appropriate given the dual nature of the investigation: developing an IoT-based irrigation monitoring system and systematically evaluating its technical performance and user acceptability. The descriptive component quantitatively characterizes the current irrigation practices and challenges, the stages of system development, the hardware effectiveness, and the expected output, while the developmental component involves the systematic creation and iterative refinement of the ArvaCore prototype using the prototyping method.

The research design follows the three-phase structure established in the conceptual framework. The Input Phase establishes the study's foundational requirements and evaluation metrics, including hardware and software components, irrigation thresholds, and Technology Acceptance Model (TAM) evaluation criteria. The Process Phase operationalizes these inputs through system prototyping and assembly, sensor data logging with error computation, and the administration of structured Likert-scale surveys. The Output Phase delivers the validated system with

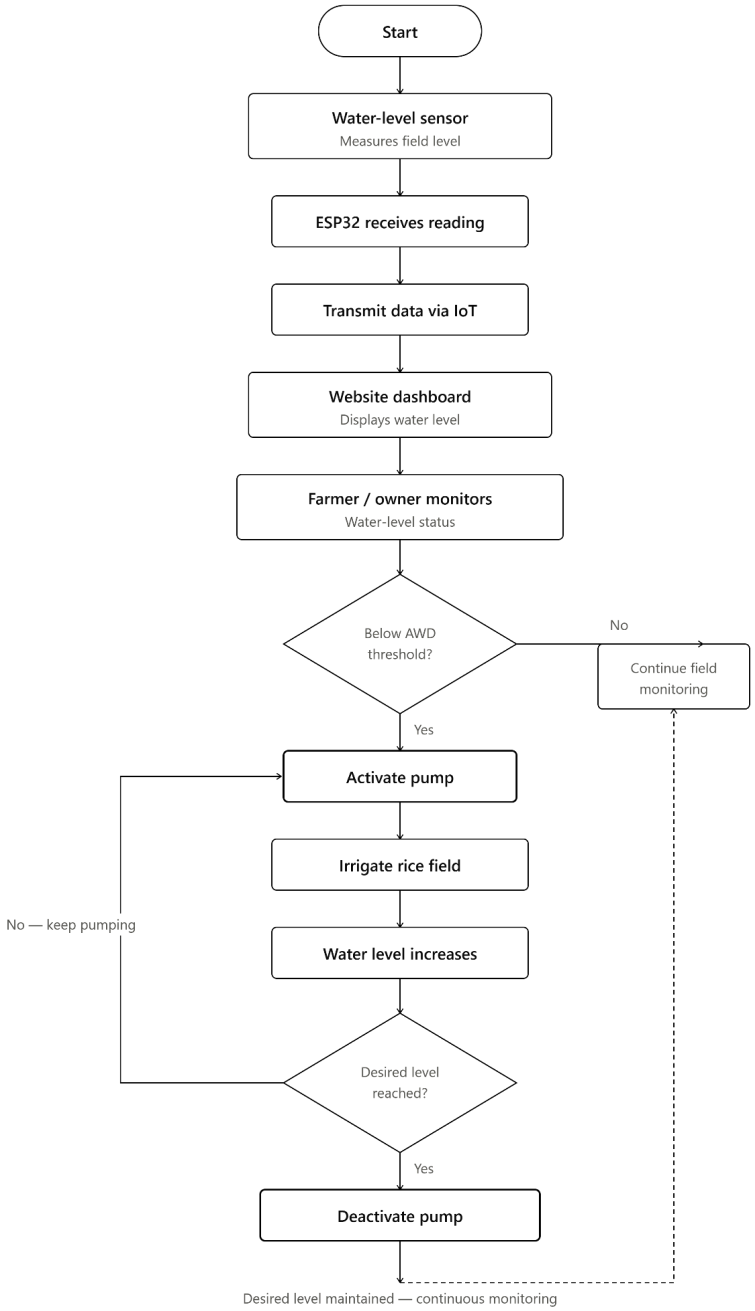
comprehensive statistical acceptability reports confirming its operational and user-centered performance. This phased structure ensures that the development of the system and the evaluation of its performance and acceptability proceed in a logical, sequential, and measurable manner.

The quantitative approach is applied consistently across all research questions. Current irrigation practices and challenges are quantified through a structured Problem Validation Report, system development stages are documented through structured developmental metrics, hardware effectiveness is measured through sensor accuracy and control performance against predetermined benchmarks, and user acceptability and expected output are assessed through a structured survey questionnaire analyzed using descriptive statistics. This unified quantitative framework ensures that the study systematically and rigorously examines both the system's technical effectiveness and its acceptance among intended users.

Flowchart of Research Design/Process Flowchart

Figure 2.

ArvaCore Process Flowchart



Description of Research Instrument Used

The study uses three research instruments: the ArvaCore prototype, the Problem Validation Report, and a structured survey questionnaire. Each instrument corresponds to specific research questions and is designed to collect quantitative data consistent with the study's descriptive-developmental research design.

The ArvaCore prototype serves as the primary technical instrument for evaluating the hardware effectiveness of the system. It is an IoT-based irrigation monitoring and water management system designed for rice cultivation, integrating distributed sensor nodes, a gateway, a cloud-based application server, a web-based monitoring dashboard, a mobile application, and a centralized automated irrigation control mechanism. Through the prototype, quantitative technical data are collected to answer the research questions on hardware effectiveness.

Water-level monitoring accuracy is assessed by comparing the readings from the ultrasonic water-level sensor with manual field measurements taken using a ruler or measuring stick, while soil-moisture monitoring accuracy is assessed by comparing the readings from the capacitive soil-moisture sensor with standard reference measurements such as gravimetric soil moisture determination. Irrigation activation accuracy measures the system's ability to initiate water release when the threshold is unmet, and irrigation deactivation accuracy measures the system's ability to shut off irrigation when the optimal water level is reached. Irrigation response time is measured as the time elapsed from the triggering of a threshold condition to the actual activation or deactivation of the irrigation system. Water consumption monitoring is assessed by

comparing the water-flow sensor readings with actual water volume measurements. All hardware performance data are collected through repeated trials under controlled field conditions and analyzed using descriptive statistics against predetermined benchmarks.

The Problem Validation Report serves as the contextual-diagnostic instrument for answering the research question on current irrigation practices and challenges. It is a structured quantitative document obtained from a farm owner in Barangay Silang, Lopez, Quezon, who is a direct beneficiary of the proposed ArvaCore system. The report is organized into a quantifiable checklist covering current irrigation practices, frequency of practice, challenges encountered, severity of each challenge measured on a 5-point Likert scale, and perceived impact on yield and water use. The data from the Problem Validation Report are coded numerically and analyzed using frequencies and percentages, weighted mean and standard deviation, and ranking. This instrument ensures that the contextual question is answered quantitatively while remaining grounded in the actual conditions of the target users.

The structured survey questionnaire is the primary user-centered instrument for answering the research questions on acceptability and expected output. It is developed based on the Technology Acceptance Model (Davis, 1989) and is designed to assess the level of acceptability of the ArvaCore system among rice farmers and agricultural practitioners, as well as their perceptions of the system's expected output.

The questionnaire consists of three sections. Section A collects demographic information about the respondents, including age, sex, educational attainment, years of farming experience, farm size, and current irrigation practices. Section B assesses the respondents' perceptions of the system's acceptability in terms of functionality, reliability, usability, and efficiency. Functionality refers to the extent to which the system performs its intended functions effectively, assessing perceptions of the system's ability to monitor field conditions, control irrigation operations, and provide relevant information. Reliability refers to the consistency and dependability of the system's performance, assessing perceptions of accurate and consistent readings, responsive commands, and operational stability. Usability refers to how easily users can learn and use the system, assessing perceptions of user-friendliness, operational simplicity, and ease of understanding. Efficiency refers to the system's ability to help users achieve their irrigation management goals with minimal effort and resource consumption, assessing perceptions of improved irrigation practices, reduced water waste, and informed decision-making.

Section C assesses the respondents' perceptions of the expected output of the study, including the system's contribution to validated irrigation monitoring, usefulness of generated data for decision-making, recommendability for adoption by other farmers, contribution to sustainable water management, and ability to meet the needs of rice farmers.

The questionnaire uses a 5-point Likert scale with the following response options and corresponding weights: Strongly Disagree (1), Disagree (2), Neutral (3),

Agree (4), and Strongly Agree (5). The questionnaire is administered to the target respondents after pilot testing to assess clarity and comprehensibility.

Population, Sample Size, and Sampling Technique

The study population consists of 3,418 rice farmers in the Municipality of Lopez, Quezon. Although the system's implementation and field testing are limited to Barangay Silang, the municipality-wide population is used because a farmer's residence does not necessarily match the location of the agricultural land they cultivate or manage. Farmers residing in other barangays may still have direct experience with rice farming and irrigation activities within the study area. The use of the broader municipality-level population therefore provides a more inclusive basis for identifying potential respondents with relevant rice-farming experience.

The sample size is determined using Cochran's sample-size procedure with a 95% confidence level, 5% margin of error, and an estimated population proportion of 0.50. The use of 0.50 is appropriate when no prior estimate of the population proportion is available because it produces the most conservative sample-size requirement. For a known finite population, the resulting initial sample size is adjusted using the finite population correction. Based on a population of 3,418 rice farmers, the resulting sample size is 346 respondents. These respondents will complete the survey questionnaire used to evaluate user acceptability and expected output of the proposed system.

The researchers will select respondents through purposive sampling, a non-probability sampling technique in which participants are deliberately chosen because they possess characteristics relevant to the study's objectives. This approach is appropriate when researchers need participants with specific knowledge, experience, or characteristics relevant to the research objectives (Nikolopoulou, 2022). Accordingly, respondents will be required to have direct involvement in rice-farming activities and relevant experience in rice cultivation, irrigation, or farm water management. Farmers or farm workers residing in other barangays may also be included if they actively cultivate, work on, or manage rice farms within the study area. This criterion-based selection aims to ensure participants can provide informed assessments of ArvaCore based on their actual experience with rice production and agricultural water management.

Statistical Treatment

The statistical treatment of the data is designed to address the research questions established in Chapter 1. Technical data are collected directly from the prototype and the Problem Validation Report, while user acceptability and expected output data are obtained through the questionnaire described in Section 3.4.

Current Irrigation Practices and Challenges (SOP 1) — The current irrigation practices and challenges encountered by rice farmers in Barangay Silang, Lopez, Quezon are analyzed using data from the Problem Validation Report. The report is structured as a

quantifiable checklist covering irrigation practices, frequency of practice, challenges encountered, severity of each challenge, and perceived impact on yield and water use.

Frequencies and percentages are used to describe the prevalence of each irrigation practice and challenge. The weighted mean and standard deviation are used to describe the severity of each challenge based on a 5-point Likert scale. Ranking is applied to identify the most to least serious challenges. This analysis provides a quantitative description of the contextual conditions that justify the proposed system.

Technical Data Analysis

Water-Level and Soil-Moisture Accuracy (SOP 3.1 and 3.2) — The accuracy of the water-level and soil-moisture sensors is assessed by comparing automated sensor readings with corresponding reference measurements obtained from the same monitoring locations and trials.

The measurement error is calculated as:

Error = Automated Reading − Reference Reading

The percentage error is calculated as:

Percentage Error = (Error / Reference Reading) × 100

The mean error, mean percentage error, and standard deviation are used to summarize the accuracy and consistency of the sensor readings. System accuracy is further expressed as the percentage of readings that fall within the predetermined

acceptable error tolerance of ±5%. Automated readings are compared directly against reference measurements to describe the magnitude and direction of any observed differences.

Irrigation Activation and Deactivation Accuracy (SOP 3.1 and 3.2) - The accuracy of the irrigation control functions is determined based on whether the system correctly activates and deactivates the irrigation mechanism when the predetermined water-level or soil-moisture thresholds are reached.

Activation and deactivation accuracy are calculated as:

- **Activation Accuracy = (Successful Activations / Total Activation Attempts) × 100**
- **Deactivation Accuracy = (Successful Deactivations / Total Deactivation Attempts) × 100**

The system is considered to meet the predetermined performance standard when the observed accuracy is at least 95% across 20 trials. Activation and deactivation accuracy are reported as descriptive percentages based on the proportion of successful operations.

Irrigation Response Time (SOP 3.3) — Irrigation response time is measured as the elapsed time, in seconds, between the occurrence of the predetermined threshold condition and the actual activation or deactivation of the irrigation mechanism. The mean and standard deviation of response time are calculated across 20 trials to

describe the system's average response and consistency. The observed mean response time is then compared against the predetermined acceptable response-time standard of 30 seconds. Results are reported descriptively, indicating whether the system's average response time falls within, meets, or exceeds the established benchmark.

Water Consumption Monitoring (SOP 3.6) - The accuracy of water-consumption monitoring is assessed by comparing the volume recorded by the flow sensor with the actual volume measured using a calibrated reference water meter.

The mean absolute percentage error (MAPE) is calculated across 10 irrigation events using:

MAPE = (1/n) × Σ |(Actual − Recorded) / Actual| × 100

The MAPE is used to describe the overall magnitude of error in water-consumption measurements, with lower values indicating greater measurement accuracy.

User Acceptability Data Analysis

Level of Acceptability (SOP 4) — The demographic profile of the respondents is described using frequencies and percentages. The responses to the user-acceptability questionnaire are summarized using the weighted mean and standard deviation.

The weighted mean is calculated using:

Weighted Mean = Σ(Item Rating × Frequency) / Total Number of Respondents

The resulting weighted means are interpreted according to the following scale:

Table 1

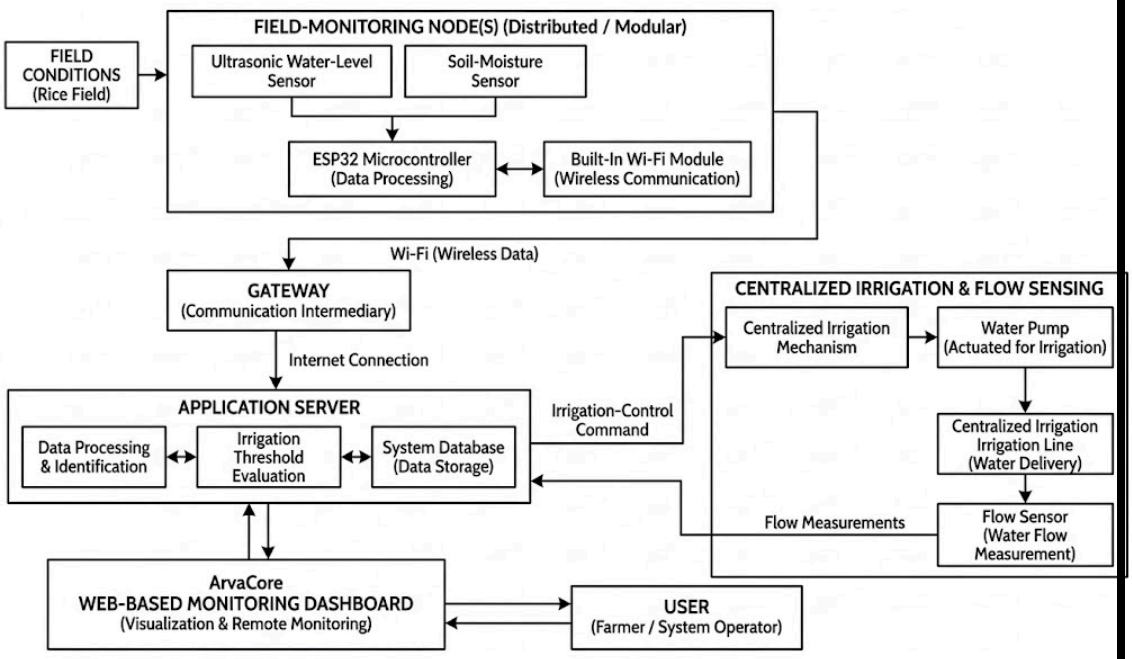
Interpretation of Weighted Means for User Acceptability Questionnaire.

Weighted Mean	Interpretation
1.00–1.80	Very Low Acceptability
1.81–2.60	Low Acceptability
2.61–3.40	Moderate Acceptability
3.41–4.20	High Acceptability
4.21–5.00	Very High Acceptability

Expected Output of the Study (SOP 5) — The responses to the expected output questionnaire are summarized using the weighted mean and standard deviation. The weighted mean is calculated using the same formula as the acceptability analysis. The resulting weighted means are interpreted using the same scale presented in Table 1. This provides a quantitative description of the respondents' perceptions of the system's expected contribution and readiness for adoption.

Design Project Flow

Figure 3.
ArvaCore System Block Diagram



The ArvaCore system consists of multiple modular field-monitoring nodes deployed at designated locations within the rice field to continuously monitor field water conditions for irrigation management. Each monitoring node consists of an ESP32 microcontroller, an ultrasonic water-level sensor, a soil-moisture sensor, and the ESP32's built-in Wi-Fi module for wireless communication. The monitoring nodes operate independently and collect water-level and soil-moisture measurements from their respective locations within the rice field, enabling the system to obtain distributed field-condition information. The number of monitoring nodes may be adjusted depending on the size and monitoring requirements of the rice field.

During system operation, each ultrasonic water-level sensor measures the distance between the sensor and the water surface by emitting ultrasonic waves and calculating the time taken for the reflected waves to return, which is then converted into the current water level at its designated monitoring location. The soil-moisture sensor measures the volumetric water content in the soil at the root zone. Both sensors provide their measurements to the corresponding ESP32 microcontroller, which processes the collected readings and transmits them through its built-in Wi-Fi module to the gateway.

The gateway serves as the primary communication intermediary between the field-monitoring nodes and the application server. It receives sensor data from the ESP32-based monitoring nodes and forwards the collected information through an Internet connection to the application server, establishing the communication link between the agricultural field and the server infrastructure.

The application server receives the data forwarded by the gateway and performs the system's data processing, storage, monitoring, and irrigation-management functions. It identifies the received measurements by their respective monitoring nodes, stores them in the system database, and makes the resulting information available through the ArvaCore web-based monitoring dashboard. The dashboard presents the current water-level and soil-moisture conditions reported by the field-monitoring nodes, enabling the farmer or system operator to remotely observe rice-field conditions and monitor changes over time.

The application server evaluates the collected water-level and soil-moisture measurements against the predefined irrigation thresholds to determine whether irrigation is required. When the monitored condition reaches the predefined threshold, the application server generates an irrigation-control command for the centralized irrigation mechanism.

In addition to the distributed monitoring nodes, the ArvaCore system includes a flow sensor in the centralized irrigation line. When irrigation is required, the centralized irrigation-control mechanism activates the water pump, and the flow sensor measures the water passing through the irrigation line. These measurements allow the system to verify that water is flowing when the pump is activated, distinguish between pump activation and actual water flow, and identify conditions such as insufficient or interrupted water delivery.

While the pump operates, the field-monitoring nodes continue measuring water levels and soil-moisture conditions at their respective locations. The updated measurements are transmitted to the gateway and forwarded to the application server, allowing the system to continuously monitor the effect of irrigation on field conditions. Once the required water condition has been reached, the irrigation-control mechanism deactivates the water pump, and the flow sensor confirms that water delivery has stopped.

When the monitored conditions do not require irrigation, the water pump remains inactive while the monitoring nodes continue collecting and transmitting

measurements. The application server processes the latest readings and displays them on the web-based monitoring dashboard for remote monitoring. The flow sensor also monitors the irrigation line during this period to verify that no unintended water flow occurs.

This monitoring and irrigation process continuously repeats throughout system operation. Distributed monitoring nodes detect changes in field-water and soil conditions resulting from crop water consumption, evaporation, drainage, rainfall, and irrigation. Throughout the whole operation, ArvaCore provides an integrated system for managing water conditions in rice cultivation while reducing the need for frequent manual field inspection.

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Appendix A

Research Questionnaire

Introduction

Dear Respondent,

We are students from the Polytechnic University of the Philippines, Sta. Mesa, Manila, conducting a study titled "ArvaCore: An IoT-Based Alternate Wetting and Drying (AWD) Irrigation Monitoring and Water Management System for Rice Cultivation."

The purpose of this questionnaire is to assess your level of acceptability of the ArvaCore system. Your honest feedback will be invaluable in evaluating the system's functionality, reliability, usability, and efficiency from the perspective of its intended users.

Your participation is voluntary, and we will treat all information you provide with strict confidentiality and use it only for academic research purposes. Please answer all questions to the best of your ability. There are no right or wrong answers.

Thank you for your time and participation.

Sincerely,

The Researchers

Part I: Respondent Profile

Instructions: Please answer the following questions by filling in the blanks or checking the appropriate box.

1. Age: _____ years old

2. Sex:

[ ] Male

[ ] Female

3. Highest Educational Attainment:

[ ] No formal education

[ ] Elementary level

[ ] Elementary graduate

[ ] High school level

[ ] High school graduate

[ ] College level / Vocational

[ ] College graduate / Postgraduate

4. Years of Farming Experience: _____ years

5. Farm Size (in hectares): _____ hectares

6. Current Irrigation Practice:

[ ] Conventional (continuous flooding)

[ ] Alternate Wetting and Drying (AWD)

[ ] Other: (please specify) _____

Part II: User Acceptability of ArvaCore

Instructions: The following statements describe the ArvaCore system. Please indicate your level of agreement with each statement by checking (✓) the box that corresponds to your response.

Rating	Description
5	Strongly Agree
4	Agree
3	Neutral
2	Disagree
1	Strongly Disagree

A. Functionality

The extent to which the system provides and performs its intended functions.

Item #	Statement	5	4	3	2	1
1	The system accurately monitors the water level in the rice field.	☐	☐	☐	☐	☐
2	The system accurately monitors soil-moisture conditions in the rice field.	☐	☐	☐	☐	☐
3	The system automatically activates the water pump when irrigation is needed.	☐	☐	☐	☐	☐
4	The system automatically deactivates the water pump when it reaches the desired water level.	☐	☐	☐	☐	☐
5	The system provides the information needed to monitor and manage irrigation conditions.	☐	☐	☐	☐	☐

B. Reliability

The system's consistency, stability, and dependability during operation.

Item #	Statement	5	4	3	2	1
6	The system's water-level readings are consistent during operation.	☐	☐	☐	☐	☐
7	The system's soil-moisture readings are consistent during operation.	☐	☐	☐	☐	☐
8	The system responds reliably when activating or deactivating the water pump.	☐	☐	☐	☐	☐
9	The system operates without frequent errors or interruptions.	☐	☐	☐	☐	☐
10	The system reliably provides notifications or alerts about relevant irrigation events.	☐	☐	☐	☐	☐

C. Usability

The extent to which users can learn, understand, and operate the system with ease.

Item #	Statement	5	4	3	2	1
11	The ArvaCore system is easy to operate.	☐	☐	☐	☐	☐
12	I can learn how to use the ArvaCore system with minimal training.	☐	☐	☐	☐	☐
13	The system's monitoring interface is clear and easy to understand.	☐	☐	☐	☐	☐
14	The system makes monitoring field conditions more convenient than manual field inspection.	☐	☐	☐	☐	☐
15	The system makes it easier to manage irrigation based on real-time field conditions.	☐	☐	☐	☐	☐

D. Efficiency

The extent to which the system helps users accomplish irrigation-management tasks while reducing time, effort, and unnecessary water use.

Item	Statement	5	4	3	2	1
16	The system helps me manage irrigation more efficiently.	☐	☐	☐	☐	☐
17	The system reduces the time and effort required for manual field inspection.	☐	☐	☐	☐	☐
18	The system helps me avoid unnecessary irrigation and water use.	☐	☐	☐	☐	☐
19	The system helps me determine when irrigation is needed.	☐	☐	☐	☐	☐
20	The system helps me accomplish irrigation-management tasks more effectively.	☐	☐	☐		

Closing Page:

Thank You for Your Participation!

We sincerely appreciate the time and effort you have taken to complete this survey questionnaire. Rest assured that your data will be handled safely and kept strictly confidential in accordance with the Data Privacy Act. Your responses are invaluable to the success of our research study entitled:

"ArvaCore: An IoT-Based Alternate Wetting and Drying (AWD) Irrigation Monitoring and Water Management System for Rice Cultivation."